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## **Activity Recognition System Report**

### **Assignment 1: IMU-Based Human Activity Recognition**

#### **1. Data Collection Parameters**

##### **Sampling Frequency and Data Rate:**

The IMU data was collected at a sampling frequency of 50 Hz. This sampling rate was achieved using a 50ms delay (`time.sleep_ms(50)`) in the deployment code, ensuring consistent data capture across all activities.

##### **Total Data Collected:**

Each activity was recorded for 3 minutes, resulting in approximately 9,000 samples per activity ( $50 \text{ Hz} \times 3 \text{ minutes} \times 60 \text{ seconds} = 9,000 \text{ samples}$ ). Data was collected for 13 different activities including sitting, standing, walking, brisk walking, jogging, jumping, cycling, stair-up, stair-down, sit-stand-sit transitions, phone interaction, eating with spoon, and pick and place.

#### **2. Window Size and Overlap Configuration**

##### **Sliding Window Parameters:**

A sliding window approach was implemented for feature extraction with the following configuration:

- Window Size: 20 samples
- Step Size: 10 samples (50% overlap)
- Window Duration: 0.4 seconds (20 samples / 50 Hz)

This configuration provided a balance between temporal resolution and feature stability. The 50% overlap ensured smooth transitions between consecutive windows and reduced the risk of missing important activity patterns at window boundaries.

#### **Feature Extraction:**

For each 20-sample window, 10 time-domain features were extracted per IMU channel (ax, ay, az, gx, gy, gz): mean, minimum, maximum, standard deviation, variance, RMS, energy, median, IQR, and zero-crossing rate. Additionally, the last values from each sensor channel were appended, resulting in 66 total features per window (10 features × 6 channels + 6 last values).

### **3. Challenges During Data Collection**

#### **Sensor Placement and Stability:**

Maintaining consistent wrist-mounted sensor positioning was challenging, especially during high-movement activities like jogging and cycling. Any slight shift in sensor orientation could introduce noise or inconsistency in the accelerometer and gyroscope readings.

#### **Activity Variability:**

Maintaining natural activity patterns for the full 3-minute duration was difficult. Activities like jogging and brisk walking required sustained effort, while transitional activities like sit-stand-sit needed careful timing to capture meaningful patterns without long idle periods.

#### **Environmental Factors:**

Data collection was affected by environmental constraints such as available space for walking and jogging, staircase accessibility, and suitable cycling routes. These factors sometimes led to interruptions or variations in the activity execution.

#### **Data Quality Issues:**

Some CSV files had encoding issues that required using 'latin1' encoding during data loading. Ensuring consistent column naming and data format across all activity files was also a challenge that needed careful verification.

### **4. Key Observations**

#### **Feature Importance:**

The Decision Tree model showed that certain features had higher discriminative power for activity classification. Features derived from the y-axis acceleration (ay) and z-axis gyroscope (gz) appeared particularly important, likely because wrist movements involve significant rotation and vertical motion components.

#### **Activity Similarity:**

Some activities showed similar IMU patterns and were occasionally confused by the model. For example, walking and brisk walking, or sitting and standing during

brief moments, shared similar sensor signatures. The 50% window overlap helped capture these subtle differences better than non-overlapping windows would have.

#### **Real-time Performance:**

The deployed model showed good real-time inference capability on the Nicla Vision board. The window-based approach with 20 samples meant the system needed 0.4 seconds of data before making a prediction, which provided a reasonable balance between response time and classification accuracy.

#### **Model Deployment:**

Converting the trained Decision Tree classifier to MicroPython using m2cgen worked effectively. The generated `score()` function successfully ran on the embedded device with consistent predictions, demonstrating the feasibility of deploying machine learning models on resource-constrained hardware for real-time activity recognition.

### **5. Conclusion**

This assignment successfully demonstrated the complete pipeline of developing an IMU-based activity recognition system, from data collection at 50 Hz through feature engineering with 20-sample windows (50% overlap) to real-time deployment. The main challenges involved maintaining sensor consistency and handling activity variability, while key observations highlighted the importance of proper feature extraction and window configuration for achieving reliable classification performance.